

**A Comparative Analysis of Clinical Responses in Patients with IDH1 versus IDH2
Mutations being treated for Clonal Cytopenia of Undetermined Significance (CCUS)**

Maria Stoentcheva

Washington University

Experiential Learning Data Analytics

Dr. Dorris Scott

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Executive Summary

This comparative analysis examines the clinical activity of IDH targeted inhibitors in patients with Clonal Cytopenia of Undetermined Significance (CCUS) who have IDH1 or IDH2 mutations. This analysis focused on treatment responses among IDH1 mutated patients treated with Ivosidenib and IDH2 mutated patients treated with Enasidenib focusing attention to hematologic improvements and disease response.

Data was obtained from pilot studies conducted at Washington University investigating treatment responses to Ivosidenib and Enasidenib among patients with IDH1 and IDH2 mutated CCUS. Data was obtained from Washington University oncology electronic data capture systems, OnCore and REDCap. Multiple data sets were combined and prepared for analysis using Microsoft Excel and R. Variable names were standardized, unnecessary data sections were removed, and datasets were consolidated in preparation for analysis. Microsoft Excel was used to generate charts and visualizations to examine treatment responses and identify patterns within the study data.

Overall, treatment with IDH targeted inhibitors demonstrated clinical activity in CCUS patients with IDH1 and IDH2 mutations. Patients treated with both Ivosidenib and Enasidenib demonstrated hematologic improvements of hemoglobin, platelet and neutrophil counts. Responses were observed in both mutation groups suggesting that targeting mutant IDH pathways may represent potential for the therapeutic strategy for patients with IDH mutated CCUS.

IDH1 mutated patients treated with Ivosidenib demonstrated higher rates of hematologic improvement and longer response durations compared to IDH2 mutated patients treated with Enasidenib. However, these conclusions should be interpreted cautiously due to differences in cohort size and difference in the amount of time patients have been followed while receiving treatment. Overall, the findings support further investigation of IDH-targeted therapies in patients with IDH mutated CCUS.

Use of AI

ChatGPT was used to assist in the paper outline and the presentation and organization of the project topics.

All data collection, analysis, interpretation and conclusions were completed without the use of AI tools. AI tools were used only as support for outlining and organization of this paper and not for analysis purposes.

Introduction

Clonal cytopenia of undetermined significance (CCUS) represents an emerging area of interest in the study of early myeloid disease, especially for patients who have persistently low blood counts, but do not yet meet the criteria for a defined hematologic malignancy. As treatment options for patients with specific molecular abnormalities continue to develop, there is growing interest in determining whether targeted therapies may provide clinical benefit before progression to a more advanced disease. This project examines treatment outcomes from pilot studies at Washington University involving patients with IDH mutated CCUS who received targeted therapy. The analysis compares hematologic and clinical response among IDH1 mutated patients treated with Ivosidenib and IDH2 mutated patients treated with Enasidenib. By evaluating changes in blood cell counts, response rates and in duration of response, this study aims to characterize the potential clinical activity of IDH targeted therapy in CCUS and identify differences in treatment response between the two patient groups.

Background

Clonal cytopenia of undetermined significance (CCUS) is a hematologic condition characterized by persistent cytopenias in the presence of one or more acquired somatic mutations without meeting diagnostic criteria for a myeloid malignancy. Individuals who have persistent, unexplained low blood counts but do not meet diagnostic criteria for a hematologic malignancy, are classified as having CCUS. CCUS serves as an early warning sign for potential disease progression as it is a condition associated with a high risk of progression to myeloid neoplasms such as acute myeloid leukemia (AML) and myelodysplastic syndrome (MDS). Currently, there are no established preventative therapies for patients with a high risk of myeloid neoplasm. IDH1 and IDH2 are genes that help cells produce energy and function normally. In certain blood disorders, such as acute myeloid leukemia (AML) and myelodysplastic syndromes (MDS), these genes can become mutated. Instead of carrying out their normal functions, the mutated proteins produce a harmful substance called 2-hydroxyglutarate (2-HG). This substance prevents immature blood cells from developing into healthy, functioning blood cells. As a result, immature

cells build up in the bone marrow, normal blood cell production decreases, and the risk of developing leukemia increases. Ivosidenib is a targeted therapy that blocks the abnormal IDH1 protein and Enasidenib is a targeted therapy that blocks the abnormal IDH2 protein. By stopping these mutated proteins, it greatly reduces the production of the harmful substance 2-hydroxyglutarate (2-HG), allowing immature blood cells to develop into healthy blood cells again. Clinical studies have shown that ivosidenib is generally safe, well tolerated, and effective in treating patients with IDH1-mutated acute myeloid leukemia (AML) and same for Enasidenib in treating patients with IDH2-mutated AML. Because patients with CCUS usually have fewer genetic mutations than patients with AML or MDS, researchers believe they may respond even better to Ivosidenib and Enasidenib. Both mutations cause similar diseases but are different genes. Although IDH1 and IDH2 mutations both lead to the production of harmful molecules 2-HG, they occur in different enzymes and different parts of the cell. Because of these biological differences, patients may not respond to treatment in the same way. Since each mutation has its own targeted therapy, comparing outcomes helps determine whether one targeted therapy leads to better hematologic improvement or longer-lasting responses than the other.

Data Sources

The data analyzed for this project was obtained from pilot studies being conducted at Washington University in the Bone Marrow Transplant department of Oncology. Study data was obtained from Washington University electronic data capture systems OnCore and REDCap. The datasets contain patient and treatment information including data relevant to hematologic parameters and treatment response. Data from multiple data sets were combined to create consolidated datasets for analysis of patients with IDH1 and IDH2 mutations. The resulting data sets were used to evaluate hematologic improvements, disease responses and response durations between the two treatment groups.

Data Preparation

The data obtained from OnCore and REDCap were reviewed and prepared for analysis using Microsoft Excel and R. Multiple data sets from the pilot studies were combined to

create consolidated data sets for both the IDH1 and IDH2 patient groups. In order to clean the raw data, variable names were renamed to improve consistency and readability across the data sets and sections of the raw data that were not relevant to this analysis were removed. The prepared datasets were then organized to facilitate evaluation of treatment responses and hematological outcomes including changes in hemoglobin, platelet and absolute neutrophil counts. The final datasets were used in Microsoft Excel to generate charts and visualizations for comparison of treatment responses between the patient groups.

Methodology

Patients eligible for these studies were aged ≥ 18 years and had confirmed IDH1-mutant CCUS with at least 6 months of unexplained cytopenias (hemoglobin < 10 g/dL, platelets $< 100 \times 10^9/L$ or absolute neutrophil count $< 1.8 \times 10^9/L$). The criteria for hematologic response is at least 8 weeks in duration of a hematologic improvement (hgb increase by 1.5 g/dL or greater, platelet increase of 30 K/cumm or greater for patients starting with greater than a 20 K/cumm and increase of platelets from less than 20 K/cumm to greater than 20 K/cumm, and at least an 100% increase and an absolute increase more than 0.5 K/cumm of absolute neutrophil count). Patients who demonstrated improvement in blood counts but did not maintain the improvement for at least 8 weeks were not classified as meeting the formal methodologic response criteria. Treatment response was evaluated by calculating the proportion of patients in each cohort who met the predefined criteria for hematologic improvement. Duration of response was evaluated from the time a response was achieved through the end of the observed response, with ongoing responses identified at the time of analysis. The time to response was also evaluated. Changes in hemoglobin, platelet and neutrophil counts were assessed by comparing baseline values with post-treatment values among patients whose CCUS classification was based on the respective cytopenia. Baseline characteristics including age, sex, hemoglobin, platelet, and ANC count were summarized for each cohort. Microsoft Excel and R were used for data preparation analysis and visualization. Descriptive analyses were used to summarize cohort characteristics and treatment outcomes. Response rates were displayed as

percentages while swimmers plots were used to visualize the duration and status of individual responses. Pairing scatter plots were used to compare baseline and post-treatment blood counts for hemoglobin, platelet and neutrophil counts. Differences in cohort size, baseline characteristics, follow up duration, and availability of specific count data were considered when interpreting comparisons between both patient groups.

Results

The two cohorts were relatively similar in age, although the IDH2 cohort was slightly older with a median age of 74 compared with 66 in the IDH1 cohort. The IDH1 group had a higher proportion of male patients and a longer median follow-up period. At baseline, IDH1 patients generally had more severe cytopenias, with lower hemoglobin and ANC values compared with the IDH2 cohort. Platelet comparisons were limited because platelet data were not available for the IDH2 group.

Hematologic responses were observed in both treatment groups. In the IDH1 cohort, 85% of patients met the predefined criteria for hematologic improvement, while 15% did not. Of the patients who did not meet the formal response criteria, 32% demonstrated overall improvement in blood counts but did not maintain the improvement for the required eight-week duration. In the IDH2 cohort, 40% of patients met the criteria for hematologic improvement, while 60% did not. Among those who did not meet the formal response criteria, 65% demonstrated overall improvement in blood counts but did not sustain the response for at least eight weeks.

The duration of response differed between the two cohorts. The average duration of response was 640 days, or approximately 21 months, for patients in the IDH1 cohort, compared with 129 days, or approximately 4 months, for the patients in the IDH2 cohort. At the time of analysis, 82% of IDH1 patients with a response had an ongoing response compared to 50% of responding IDH2 patients. The average time required to achieve a response was 37 days for the IDH1 cohort, and 45 days for the IDH2 cohort.

Hematologic improvements were also observed when individual blood cell lineages were examined. Among patients who qualified for CCUS based on a low ANC, 100% of patients

in both the IDH1 and IDH2 cohorts demonstrated improvement in neutrophil counts compared with their baseline values. Hemoglobin and platelet analyses were more limited because relatively few patients qualified for CCUS based on those cytopenias. Only one patient in each cohort qualified for CCUS based on low hemoglobin counts at baseline and no IDH2 patients qualified for CCUS based on low platelet counts. Despite the limited sample sizes, all patients evaluated for hemoglobin and platelet response demonstrated improvement compared with their baseline values.

Discussion

Both patients in the IDH1 cohort and IDH2 cohort experienced hematologic improvements in disease responses. Improvements were observed across differing blood count categories of hemoglobin, and absolute neutrophil count, suggesting that targeted in addition of mutant IDH pathways may have potential to improve abnormal blood counts in patients with CCUS. Notably, 85% of patients in the IDH1 cohort met the predefined criteria for hematologic improvement compared with 40% in the IDH2 cohort. The IDH1 patient group also demonstrated a longer average duration of response approximately 640 days compared with 129 days in the IDH2 group. These findings suggest that IDH1 patients treated with Ivosidenib have higher rates of hematologic improvement and more durable responses than IDH2 patients treated with Enasidenib.

However, these differences should be interpreted with caution. The IDH1 and IDH2 cohorts differed in cohort size, Length of follow up and in the amount of time patients had been receiving study treatment. The IDH2 cohort had less accrued patience and was earlier in the treatment process making direct comparisons between the groups difficult. Baseline differences were also present, with the IDH1 cohort generally having more severe cytopenias and a longer median follow-up period. In addition, platelet response could not be meaningfully compared because platelet data was not available for the IDH2 cohort, and hemoglobin and platelet analysis were limited by the small number of patients with the specific cytopenias. These limitations make it difficult to determine whether the observed

differences in response are related to mutation type, treatment, disease characteristics, duration of treatment, or differences in cohort composition.

The significance of this analysis is particularly important because much of the existing research on targeted IDH therapies has focused on patients with acute myeloid leukemia (AML), while substantially less information is available regarding their activity in the CCUS. By examining hematologic improvement, response rates, time to response, and response duration, this study provides an early assessment of the potential role of targeted IDH inhibition in the earlier stages of myeloid disease. The observed responses in both cohorts support continued investigation of these therapies and provide preliminary data that may help guide future treatment strategies and larger prospective studies.

Conclusion

Treatment with IDH targeted inhibitors demonstrated clinical activity and patients with both IDH1 and IDH2 mutated CCUS, with both cohorts showing hematologic improvements and disease responses. Patients demonstrated improvements in affected blood values including hemoglobin platelet and your triple recovery when these measures were available for evaluation. The IDH1 cohort demonstrated a higher rate of hematologic improvement and a longer average duration of response than the IDH2 cohort however these findings should be considered preliminary due to the difference in cohort size and time on treatment. The IDH2 cohort has not yet reached the same level of enrollment and treatment duration as the IDH1 cohort. Continued enrollment and longer treatment times will provide additional data to determine whether mutation type is associated with differences in likelihood of response time to response, duration of response, and the mental logic recovery. Overall, these findings support further investigation of IDH targeted therapies in CCUS and provide preliminary data that may inform future studies a targeted treatment for patients with IDH mutations.

References

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Appendix A

Disease Response for IDH1 Patients

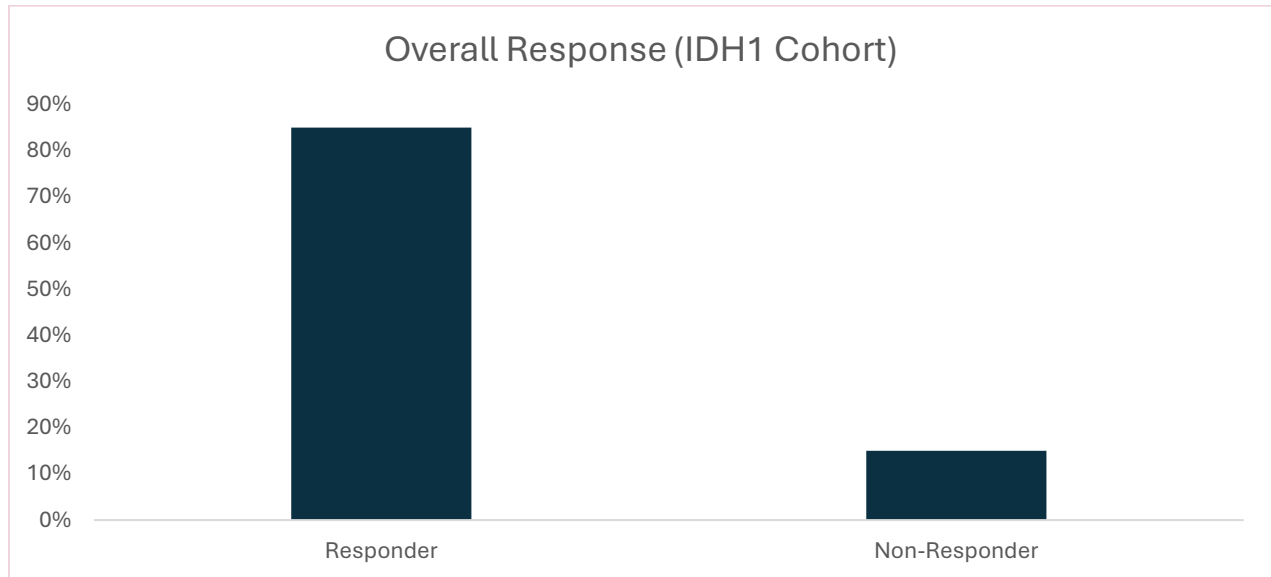


Figure A1. Percentage of IDH1 Patients who met Criteria for Disease Response.

Appendix B

Disease Responses for IDH2 Patients

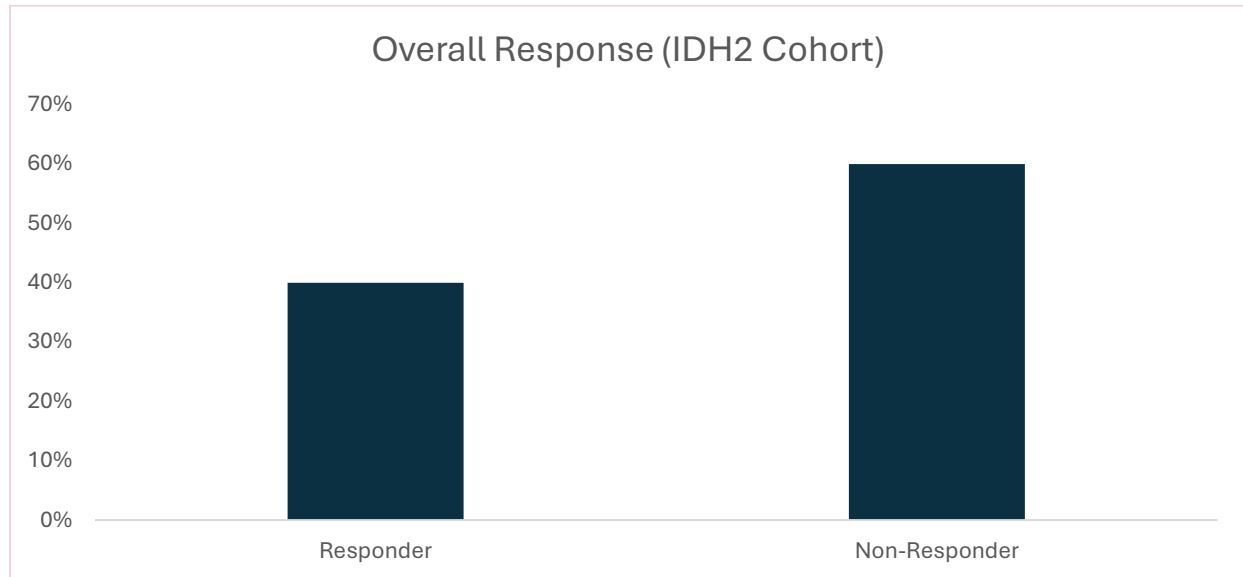


Figure B1. Percentage of IDH2 Patients who met Criteria for Disease Response.

Appendix C

Duration of Response in IDH1 Patients

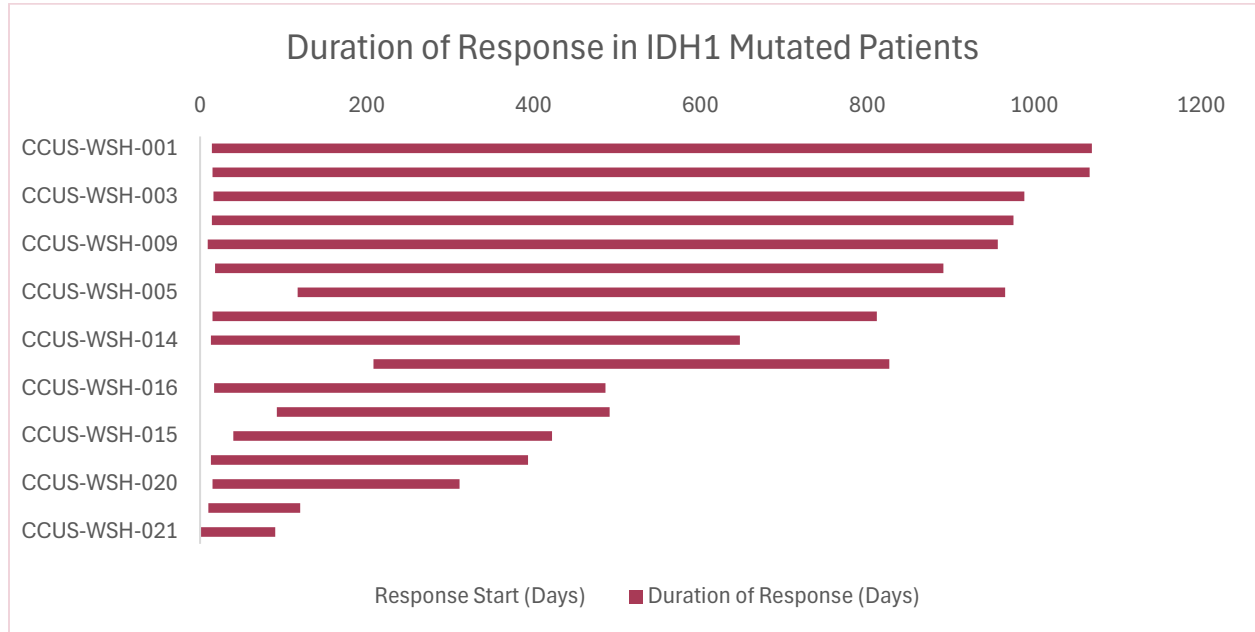


Figure C1. Duration of Response in IDH2 Patients in Days.

Appendix D

Duration of Response in IDH1 Patients

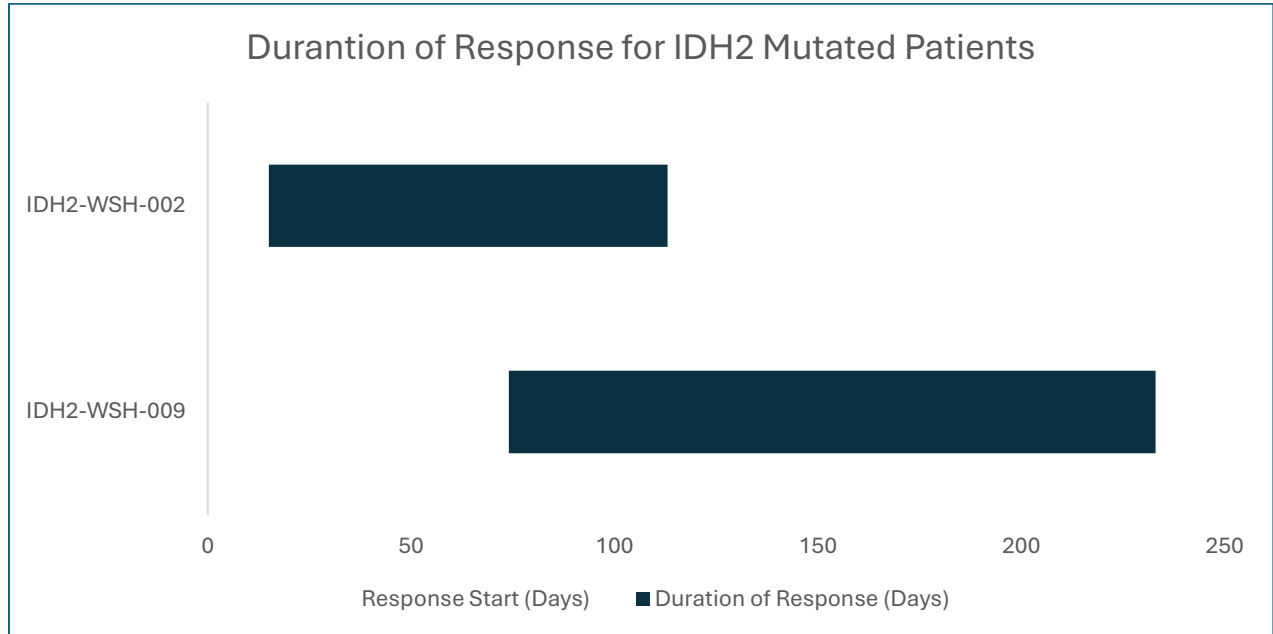


Figure D1. Duration of Response in IDH2 Patients in Days.

Appendix E

Neutrophil Count Improvement

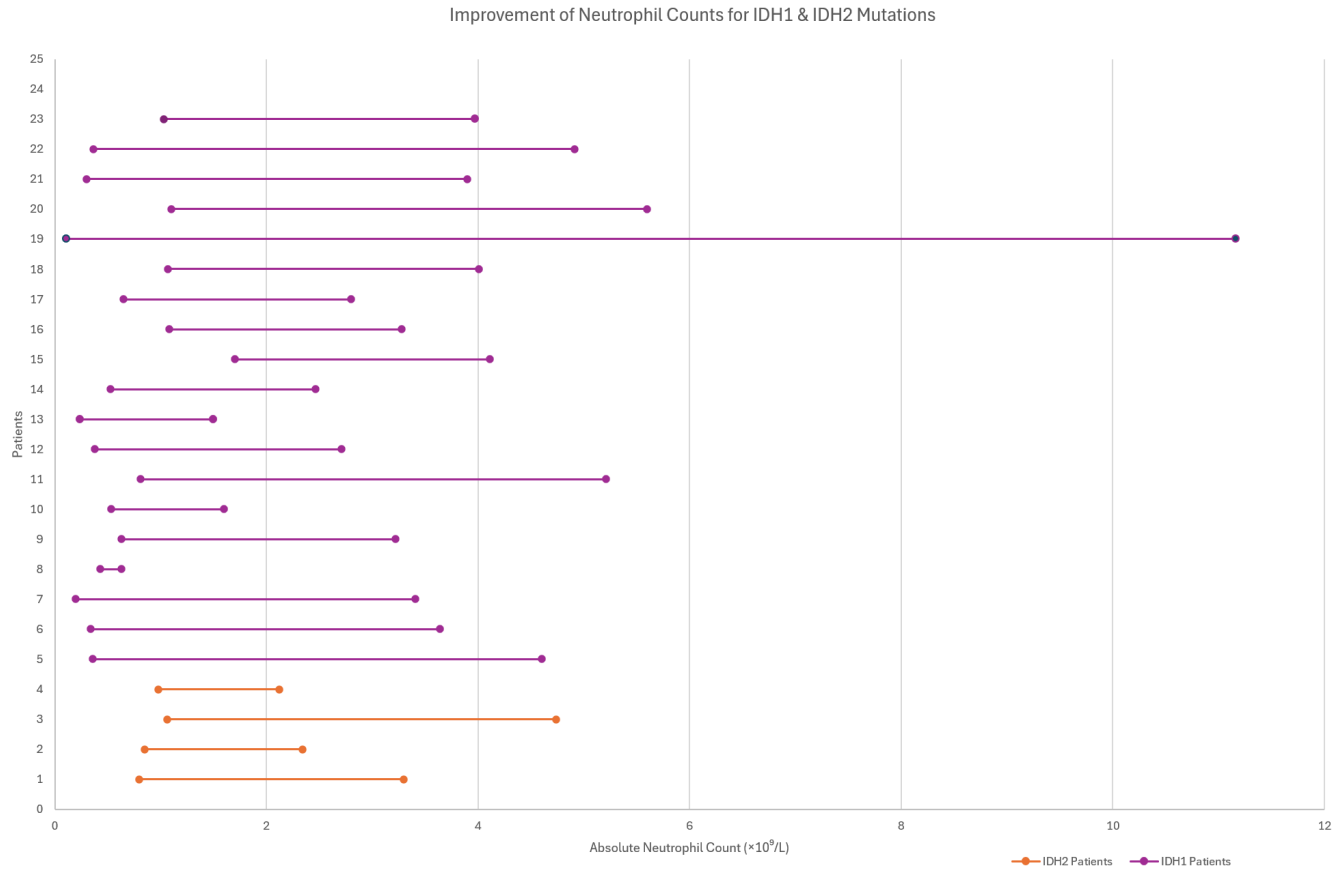


Figure E1. Neutrophil Count Improvement for IDH1 and IDH2 Mutated Patients.

Appendix F

Hemoglobin Count Improvement

Improvement of Hemoglobin Counts for IDH1 & IDH2 Mutations

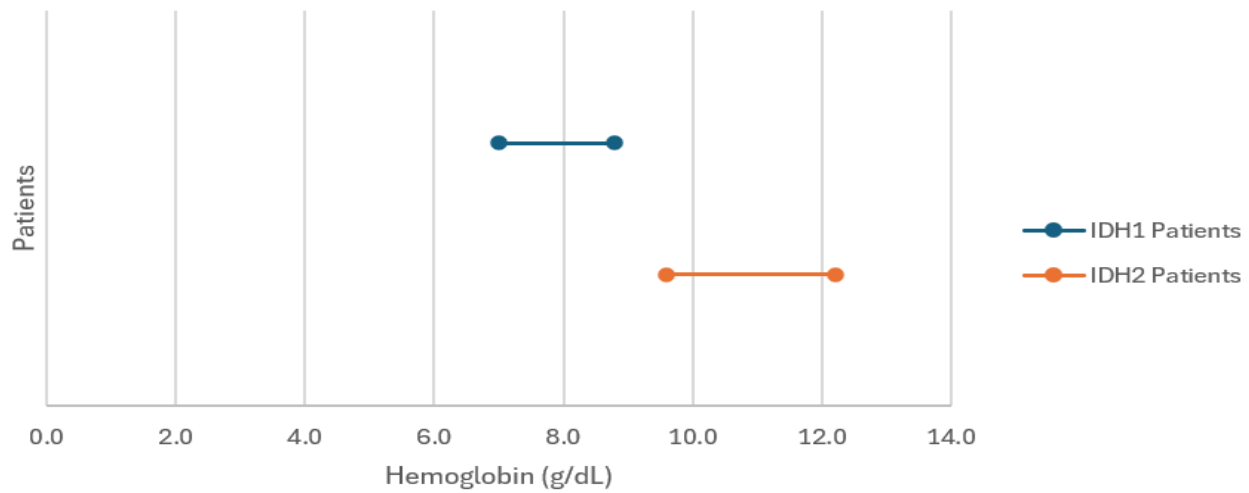


Figure F1. Hemoglobin Count Improvement for IDH1 and IDH2 Mutated Patients.

Appendix G

Platelet Count Improvement

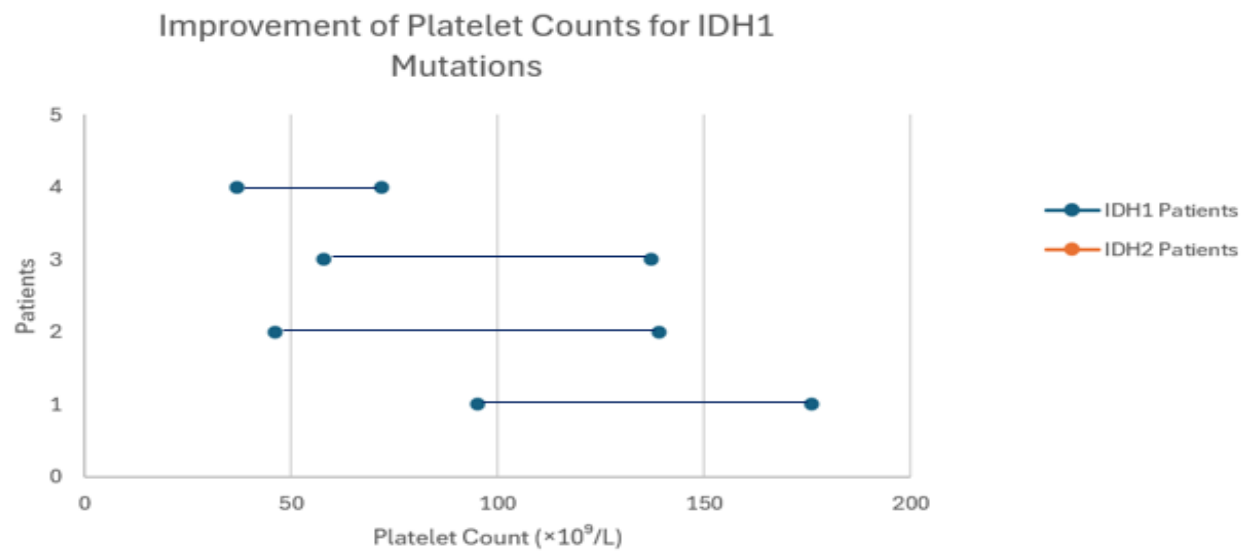


Figure G1. Platelet Count Improvement for IDH1 and IDH2 Mutated Patients.

Appendix H

Patient Demographics

	IDH1 Cohort (n=20)	IDH2 Cohort (n=5)
Median age (range)	66	74
Male sex (%)	70	20
Female sex (%)	30	80
Median follow-up (cycles)	22	9
Median baseline hemoglobin (g/dL)	7	9.6
Median baseline platelets ($\times 10^9/L$)	52	n/a
Median baseline ANC ($\times 10^9/L$)	0.527	0.915

Figure H1. IDH1 and IDH2 Patient Demographics